

Notes: Prendergast and Topel (JPE, 1996)

Favoritism in Organizations

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1 Big picture

- **Question.** How does favoritism by supervisors affect incentive pay, performance evaluation, job assignment, bureaucracy, and the allocation of authority inside organizations?
- **Core setting.** Objective measures of employee performance are often unavailable. Firms therefore rely on subjective evaluations by supervisors, but subjectivity lets supervisors favor or penalize subordinates based on personal preferences.
- **Main tension.** Supervisors value power over subordinates because their reports affect worker welfare. Firms must trade off this benefit of authority against the costs of favoritism: arbitrary rewards, extra risk imposed on workers, distorted performance information, and inefficient job assignments.
- **Key mechanism.** The more a supervisor's report matters for a worker's pay, the more the supervisor is willing to distort the evaluation. Incentive pay therefore corrupts the very signal used to implement incentive pay.
- **Why this differs from standard moral hazard.** In a standard agency model, measurement error is exogenous. In this paper, measurement error is endogenous: the compensation contract changes the amount of bias in the supervisor's report.
- **Main reward-design result.** If favoritism is costly at the margin, firms optimally weaken worker incentives and place less weight on subjective evaluations.
- **Main organizational-design result.** Favoritism can rationalize bureaucratic rules that give too little weight to supervisor appraisals and too much weight to noncorruptible measures such as seniority or formal metrics.
- **Counterintuitive result.** Favoritism is not always bad for the organization. If the firm can price supervisor power efficiently, the value supervisors place on authority can be transformed into organizational surplus, and worker incentives need not fall.
- **When favoritism becomes harmful.** Distortions in the "market for power" - especially supervisor risk aversion or the need to incentivize supervisor monitoring effort - can make favoritism costly and push the firm toward weaker incentives and more bureaucratic rules.
- **Economic takeaway.** The paper treats firms as social institutions. Because supervisors have preferences over subordinates, the design of pay and authority cannot be separated from the social relations inside the firm.

2 Incremental contribution of the paper

- **From objective performance measurement to subjective evaluation.** Much of incentive theory treats performance measures as given. This paper starts from the fact that many important performance evaluations are subjective and therefore manipulable.
- **A theory of favoritism rather than bribery.** The paper does not require explicit side payments or lobbying by workers. Bias arises because supervisors have likes and dislikes toward subordinates.

- **Endogenous measurement noise.** The model shows that stronger incentives can create more noise in performance measurement because supervisors distort more when their evaluations matter more.
- **Authority as a compensation good.** Supervisors value the ability to affect workers' welfare. The model therefore derives a benefit of being "boss" and studies how firms optimally allocate that authority.
- **Bureaucracy from incentive design.** The paper gives an economic rationale for bureaucratic rules that intentionally ignore relevant subjective information. Such rules can be optimal when subjective evaluations are corruptible.
- **Task assignment channel.** Bias matters not only because it makes pay unfair or risky, but also because it corrupts information used to place workers in the tasks where they are most productive.
- **A non-monotonic view of discretion.** Supervisor discretion can be too high or too low. If favoritism is harmful, firms restrict discretion; if supervisor power creates enough surplus, firms may give supervisor evaluations more weight.
- **A framework for group differences.** The model distinguishes expected bias from uncertainty about bias. Differences in the variance of favoritism across worker groups can change the use of subjective versus objective performance measures.
- **Link to organizational economics.** The paper connects agency theory, subjective performance evaluation, bureaucracy, job assignment, and discrimination in one unified model.

3 Model environment and primitives

3.1 Actors and timing

- The firm employs one worker and one supervisor.
- The worker chooses effort e at cost $c(e)$, with $c'(e) > 0$, $c''(e) > 0$, $c'(0) = 0$, and $c'(\infty) = \infty$.
- The supervisor privately observes a noisy, nonverifiable performance signal and reports an evaluation to the firm.
- The firm also observes its own noisy signal of performance, which is noncorruptible but imperfect.
- Worker pay and supervisor pay are set before performance is realized.
- After pay is determined, the worker may be assigned to a different task based on the firm's estimate of ability.

Interpretation: The supervisor is a source of useful information, but also a source of distortion. The firm cannot simply ignore the supervisor because the supervisor observes information that is useful for incentives and assignment.

3.2 Performance technology

The supervisor observes

$$y = e + \alpha + \varepsilon_s,$$

where e is effort, α is the worker's match-specific talent for the current task, and ε_s is supervisor measurement error.

The firm observes

$$y_f = e + \alpha + \varepsilon_f.$$

- Talent α is unknown to all parties and is normally distributed with mean zero.
- Supervisor error ε_s and firm error ε_f are normally distributed and independent of talent.
- The supervisor's signal is useful because it contains information about α .
- Talent is task-specific. A worker with positive expected α should stay in the current task; a worker with negative expected α should be reassigned.

Interpretation: The same information is used for two purposes: motivating effort and assigning workers to tasks. This is why favoritism can damage both incentives and productivity.

3.3 Worker preferences

The worker has exponential utility:

$$v = -\exp\{-r[w - c(e)]\},$$

where $r > 0$ measures absolute risk aversion.

- The worker dislikes effort cost.
- The worker dislikes wage risk.
- Any arbitrariness in subjective evaluations must be compensated by a risk premium.

Interpretation: Favoritism is costly partly because it exposes workers to risk unrelated to true performance. When supervisors' preferences are random from the worker's perspective, subjective evaluation becomes a noisy lottery.

3.4 Supervisor preferences and favoritism

The supervisor's utility is

$$v_s = w_s + \eta w,$$

where w_s is supervisor pay, w is worker pay, and η is the supervisor's personal preference for the worker.

- If $\eta > 0$, the supervisor likes the worker and wants the worker to receive higher pay.
- If $\eta < 0$, the supervisor dislikes the worker and wants the worker to receive lower pay.
- The supervisor learns η after entering the relationship with the worker.
- The ex ante distribution of η has mean zero and variance σ_η^2 .

Interpretation: Favoritism is modeled as positive or negative altruism toward the subordinate. What matters for contract design is not only average favoritism, but uncertainty about which way the supervisor will be biased.

3.5 Supervisor monitoring and worker compensation

The firm penalizes deviations between the supervisor's report y_s and the firm's own information y_f :

$$w_s = w_0 - .5\lambda(y_s - y_f)^2.$$

Worker pay is linear in the two signals:

$$w = \tau_0 + \tau_s y_s + \tau_f y_f.$$

- λ is the price of perceived favoritism. A higher λ makes it more costly for the supervisor to bias the report.
- τ_s measures how much the supervisor's report matters for worker pay.
- τ_f measures how much the firm's noncorruptible signal matters for worker pay.
- The paper studies how the firm chooses τ_s , τ_f , and λ .

Interpretation: The firm has two tools: weaken the worker's reward for supervisor reports, or increase the penalty on supervisor bias. The first tool is indirect; the second tool directly prices the supervisor's power.

4 Core mechanism: endogenous bias

4.1 Supervisor reporting rule and compression

Because the supervisor is monitored by comparison with the firm's signal, the supervisor forms an expectation of what the firm saw:

$$E(y_f | y, e^*) = \theta y + (1 - \theta)e^*,$$

where

$$\theta = \frac{\sigma_\alpha^2}{\sigma_\alpha^2 + \sigma_s^2}.$$

The supervisor reports

$$y_s = E(y_f | y, e^*) + b.$$

- The term $E(y_f | y, e^*)$ is the supervisor's best guess of the firm's information.
- The coefficient θ captures compression. If the supervisor's signal is noisy, the supervisor reports a compressed version of observed performance to avoid looking biased.
- The term b is intentional bias driven by the supervisor's personal preference.

Interpretation: Even without favoritism, monitoring can produce compressed ratings: high performers are understated and low performers are overstated. With favoritism, an additional intentional distortion appears.

4.2 Bias chosen by the supervisor

The supervisor chooses bias to maximize the personal benefit of raising or lowering the worker's pay minus the expected penalty from being detected:

$$\max_b \eta \tau_s b - \frac{\lambda}{2} b^2.$$

The first-order condition gives

$$b = \frac{\eta \tau_s}{\lambda}.$$

- Bias increases with the supervisor's preference η .
- Bias increases with τ_s , the importance of the supervisor's report for worker pay.
- Bias decreases with λ , the penalty for perceived favoritism.

Interpretation: This equation is the key to the paper. Supervisors lie more when money is on the line. Therefore, stronger incentive pay creates stronger incentives for supervisors to manipulate evaluations.

4.3 Endogenous variance of bias

Since η is random ex ante, the variance of bias is

$$\sigma_b^2 = \frac{\tau_s^2 \sigma_\eta^2}{\lambda^2}.$$

- Increasing the weight on the supervisor's report raises the variance of bias.
- Increasing the penalty on supervisor bias lowers the variance of bias.
- This variance is not an exogenous measurement-error parameter; it is generated by the compensation system.

Interpretation: The contract creates its own noise. This is why favoritism cannot be treated as ordinary signal imprecision.

5 Surplus, incentives, and bureaucracy

5.1 Components of organizational surplus

The firm's surplus maximization problem contains four conceptual components:

- Expected output net of worker effort cost: $e - c(e)$.
- Risk premium paid to the worker for noisy compensation.
- Supervisor surplus from power, because the supervisor values being able to affect worker welfare.
- Misallocation cost $L(\sigma_b^2)$ from using biased reports to assign workers to tasks.

The paper shows that greater favoritism raises misallocation costs:

$$L'(\sigma_b^2) > 0.$$

Interpretation: Bias is costly because it corrupts the information that tells the firm whether the worker fits the current task. A biased supervisor may keep the wrong worker or reassign the right worker.

5.2 When is favoritism costly?

The paper summarizes the marginal value of favoritism using a term Ω . In simplified words:

$$\Omega = \text{marginal worker risk cost} + \text{marginal misallocation cost} - \text{marginal supervisor power benefit}.$$

- If $\Omega > 0$, additional favoritism reduces total surplus.
- If $\Omega < 0$, additional favoritism raises total surplus because the supervisor's value of power is large relative to the costs.
- This sign determines whether the firm should restrict or accommodate supervisor discretion.

Interpretation: The key question is not whether favoritism exists, but whether the marginal costs of favoritism exceed the value supervisors place on authority.

5.3 Optimal worker incentives

The worker's effort satisfies

$$c'(e) = \text{effective incentive from the supervisor signal} + \text{incentive from the firm's signal}.$$

The optimal contract responds to favoritism in two ways:

- Favoritism adds risk to pay, which weakens incentives for a risk-averse worker.
- Favoritism makes the supervisor's report more corruptible when τ_s is high, which further weakens incentives if $\Omega > 0$.
- If $\Omega < 0$, the firm may strengthen incentives because doing so gives supervisors more valued authority.

Interpretation: The standard lesson is that noisier performance measures receive less weight. The new lesson is that the performance measure becomes noisier precisely when the firm gives it more weight.

5.4 Bureaucracy as inefficient information aggregation

The paper defines bureaucracy as a decision rule in which information is not aggregated in the ex post efficient way. In this model, bureaucracy appears when the ratio of weight on the supervisor's report to weight on the firm's signal differs from the ratio that would be optimal if both were simply noisy but honest signals.

- If favoritism is harmful at the margin, the firm places too little weight on supervisor reports.
- The firm places too much weight on noncorruptible signals, even if those signals are less informative in a statistical sense.
- Seniority rules, formal metrics, external review committees, or rigid promotion procedures can be interpreted as ways to reduce discretion over rents.

- If favoritism is valuable at the margin, the opposite form of bureaucracy can occur: the supervisor's report can receive too much weight because the firm accommodates the supervisor's demand for power.

Interpretation: Bureaucracy is not just red tape. It can be an optimal commitment device that protects the organization from biased subjective evaluations.

6 Monitoring the supervisor and the market for power

6.1 Selling the right to exert bias

The firm can choose λ , the price supervisors pay for perceived favoritism. If this price can be set optimally, the firm effectively sells supervisors the right to exercise power.

- Without task-misallocation costs, efficiently priced favoritism does not reduce worker incentives.
- Worker pay rises to compensate for the extra risk created by favoritism.
- Supervisor surplus from power offsets the worker risk premium.
- With task-misallocation costs, the firm may actually increase the weight on supervisor reports to create more valued authority, while charging a higher price for bias.

Interpretation: The paper's counterintuitive lesson is that supervisor power can be part of the compensation package. If priced efficiently, it can create surplus instead of merely destroying it.

6.2 Supervisor risk aversion

When supervisors are risk averse, the market for power is distorted.

- Raising λ has two opposing effects.
- It increases the variance of supervisor pay conditional on a given discrepancy between reports.
- It also reduces chosen bias, which can lower the variance of supervisor pay.
- If monitoring is poor, the first effect dominates, and the firm sets a lower λ than the efficient price.
- The result can be weaker worker incentives to reduce the scope for costly favoritism.

Interpretation: Risk aversion can make the firm unable to charge the correct price for power. When that happens, favoritism becomes inefficient and the firm may restrain worker incentives instead.

6.3 Supervisor monitoring effort

The paper also studies a case where the supervisor can exert effort to improve the precision of his observation.

- A higher λ affects both bias and supervisor monitoring effort.
- The firm wants λ to provide incentives for accurate monitoring, but the same λ also controls favoritism.

- One instrument must perform two tasks, so the market for favoritism is distorted.
- If supervisor effort is too low at the first-best bias level, the firm raises λ and may also increase worker incentives, giving the supervisor more valuable authority and inducing more effort.
- If monitoring effort is too high and bias is harmful, the firm lowers λ and weakens worker incentives to limit bias.

Interpretation: Monitoring supervisors is itself an agency problem. The firm cannot separately price bias and motivate supervisor information gathering, so second-best distortions arise.

7 Extensions and empirical content

7.1 Compression of evaluations

- Compression occurs when supervisors report evaluations closer to the mean than their true observations.
- In the model, compression arises because supervisors are monitored by comparing their reports to the firm's own signal.
- If the supervisor gives an extreme rating that differs from the firm's signal, he risks being interpreted as biased.
- When favoritism is harmful, compression lowers welfare and weakens incentives because the firm must raise the piece rate on compressed reports to achieve the same incentive effect, which increases bias.
- If the firm can choose the price of bias optimally and there are no sorting costs, compression need not change worker incentives.

Interpretation: Rating compression is not simply a psychological mistake. It can be an optimal response by a supervisor who wants to avoid looking biased.

7.2 Residual claimancy

- A possible way to control favoritism is to make the supervisor a residual claimant on the worker's output.
- Residual claimancy gives the supervisor a direct cost of arbitrary evaluations because bad assignments reduce output.
- However, this solution can create new problems: the supervisor may renege on promised wages, and reputational concerns can create their own distortions.
- For example, a supervisor who fears being labeled discriminatory may inflate evaluations of minority workers even when true performance is weak.

Interpretation: Giving supervisors stronger ownership-like incentives does not automatically solve favoritism. It changes the supervisor's incentives, but can introduce new distortions.

7.3 Observable worker characteristics

The paper distinguishes two cases:

- Differences in the mean of supervisor preferences across groups.
- Differences in the variance of supervisor preferences across groups.
- Under continuous signals, expected favoritism has little effect on pay or incentives in the model's benchmark environment.
- Uncertainty about favoritism matters more. If supervisors' preferences toward a group are more variable, subjective evaluations impose more risk and create more allocation errors.
- If favoritism is harmful, the firm responds by reducing the weight on subjective evaluations for that group and increasing reliance on explicit, noncorruptible measures.
- This prediction is connected to evidence that women are more likely to work in occupations relying on objective performance measures.

Interpretation: The model does not say only that discrimination changes average outcomes. It also says that uncertainty about bias changes the structure of compensation and evaluation systems.

8 Figures

8.1 Figure 1: Input Possibility Frontier between Supervisor Bias and Monitoring Effort; Figure 2: Shift of the Bias-Effort Frontier from Stronger Worker Incentives

Read:

- **Figure 1** plots the input possibility frontier relating bias σ_b to supervisor effort e_s .
- Moving along the frontier, higher λ induces more supervisor effort and lower bias.
- Point B represents the first-best level of bias σ_b^* with supervisor effort $\hat{e}_s$.
- If effort is too low relative to the efficient effort level, the firm can increase λ , reduce bias, and raise monitoring effort.
- **Figure 2** shows how increasing worker incentives shifts the input possibility frontier outward.
- When worker incentives are stronger, the supervisor's report matters more for pay, so favoritism becomes more valuable and the bias-effort tradeoff shifts.
- The movement from D to E illustrates a case in which the firm increases worker incentives, generates more supervisor effort, and accepts more bias.

Economic message: The two figures visualize the central second-best problem. The firm uses worker incentives and supervisor monitoring penalties to jointly control bias and information gathering. Whether this leads to more or less worker incentive pay depends on whether the supervisor's power is valuable enough to offset the costs of bias.

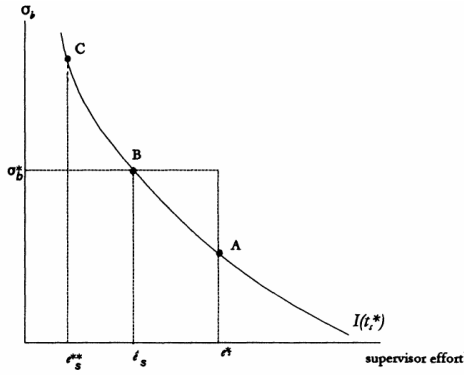


FIG. 1

Figure 1: Input possibility frontier between supervisor bias and monitoring effort.

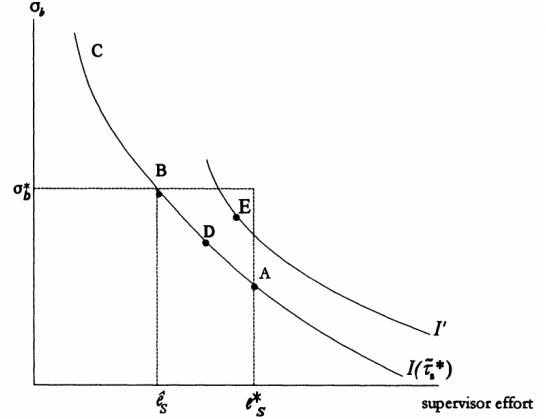


FIG. 2

Figure 2: Shift of the bias-effort frontier from stronger worker incentives.

9 Short summary

- Performance evaluation is often subjective, and subjectivity gives supervisors scope for favoritism.
- Supervisors distort reports more when their reports have larger effects on worker pay.
- Incentive contracts therefore affect not only worker effort, but also supervisor reporting behavior.
- Favoritism imposes costs through worker risk and distorted task assignment.
- Favoritism also creates benefits because supervisors value authority over subordinates.
- If the firm can price supervisor power efficiently, favoritism need not weaken worker incentives.
- If the market for power is distorted by supervisor risk aversion or supervisor monitoring effort, favoritism can become harmful and lead to low-powered incentives.
- Bureaucratic rules can be optimal because they commit the organization to limit reliance on corruptible subjective evaluations.
- Groups that face more variable potential bias may be placed in jobs or compensation systems that rely more on objective performance measures.

10 Conclusion

10.1 Core conclusion

Prendergast and Topel show that favoritism is an inherent issue in organizations that use subjective performance evaluations. The problem is not simply that supervisors may be biased; it is that the firm's compensation system determines how much bias supervisors choose to exercise.

10.2 Economic intuition

When the supervisor's report strongly affects the worker's wage, a supervisor who likes the worker gains more from inflating the report, while a supervisor who dislikes the worker gains more from deflating it. The firm therefore faces a tradeoff: subjective evaluations contain useful information, but making them consequential also makes them more corruptible.

10.3 Incremental contribution revisited

The paper expands agency theory by making subjective evaluation endogenous. It explains why firms may weaken incentives, rely on seniority or objective metrics, compress ratings, and create bureaucratic review systems. These practices may look inefficient *ex post*, but they can be optimal *ex ante* because they constrain favoritism.

10.4 Implications for organizational design

- Firms should use subjective evaluation carefully when evaluations distribute rents.
- Bureaucratic rules can protect against biased evaluation but may discard useful information.
- Objective performance measures are especially valuable when supervisor bias is likely or hard to price.
- Authority is a compensation good for supervisors, so restricting authority may require compensating supervisors in other ways.
- Organizations should recognize that changing worker incentive pay can change supervisor behavior, not just worker behavior.

10.5 Limitations

- The model is theoretical and does not estimate the magnitude of favoritism in real organizations.
- The framework uses a stylized one-worker, one-supervisor setting.
- The model assumes normally distributed signals and linear compensation to obtain tractable closed-form results.
- The paper highlights mechanisms rather than providing direct empirical tests.
- It abstracts from many organizational realities such as teams, multiple supervisors, explicit politics, repeated relationships, and external labor-market discipline.

10.6 Takeaway

The paper's main lesson is that incentive systems are social systems. Firms do not only choose pay-for-performance rules to motivate workers; they also choose how much power to give supervisors whose subjective evaluations can be biased. Favoritism can justify low-powered incentives and bureaucracy, but it can also explain why organizations grant authority to bosses who value control.